PREDICTABLE DESIGNS

SWAP LIST PDF

PRODUCTION-GRADE SENSOR SWAP LIST

Production-Grade Sensor Swap List

Swap these seven hobbyist sensors before your design goes into production.

Every sensor in this guide works fine on a breadboard, and that is exactly what makes each one dangerous. A part that passes every test on your bench can still fail once it is inside a manufactured product, where it gets judged on accuracy, lifecycle, compliance, power, and safety. The seven sensors below fail at least one of those tests, and some of them fail several at once.

Each entry explains why the sensor breaks down in production and gives you replacement part numbers that were actually designed to go inside products, usually for close to the same cost in volume. The list follows the same countdown as the video, starting with a sensor that is simply inaccurate and ending with one that is genuinely dangerous.

The Swap List at a Glance

#	IF YOUR DESIGN USES	START YOUR SEARCH WITH	THE CORE PROBLEM
7	DHT11 / DHT22	**Sensirion SHT40 or SHT41**	Accuracy and consistency
6	HC-SR04 ultrasonic module	**ST VL53 time-of-flight family**	Integration and drift
5	HC-SR501 PIR module	**Panasonic EKMC / EKMB series**	Production repeatability
4	MQ-2 / MQ-135 gas sensors	**Sensirion SGP40 / SCD40**	Power and calibration
3	MPU-6050 IMU	**TDK ICM-42688-P**	Discontinued part
2	CdS photoresistor (LDR)	**Vishay VEML7700**	RoHS compliance
1	ACS712 module on mains	**Shunt + TI AMC1302 isolated amp**	Shock and fire safety

Every replacement above has alternates listed in its full entry, so keep reading before you lock in a part.

#7 DHT11 and DHT22

TEMPERATURE AND HUMIDITY

WHY IT FAILS IN A PRODUCT

- Accuracy is only rated to ±2°C, the humidity readings are even worse, and two sensors from the same batch will noticeably disagree.
- The nonstandard one-wire protocol has strict timing requirements, so readings randomly get corrupted.
- These parts are cranked out by countless factories of unknown origin, so you never really know what you are getting, and the DHT22 shares the same protocol, variance, and supply chain problems.
- In a product, all of that adds up to bad data, support tickets, and returns.

SWAP IT FOR

REPLACEMENT PART	WHY THIS PART
Sensirion SHT40 / SHT41	These tiny sensors are factory calibrated, they communicate over standard I^2C, and their accuracy is specified across the full operating range.
TI HDC3020	This part is another fully specified option from a major manufacturer.
Bosch BME280	This part adds barometric pressure sensing in the same package.

Cost check

In production volumes you are talking a couple of dollars, which is a small price to pay for temperature and humidity data you can trust.

#6 HC-SR04 Ultrasonic Module

DISTANCE SENSING

WHY IT FAILS IN A PRODUCT

- This is a 5V module living in a 3.3V world, so you need level shifting just to connect it safely to most modern microcontrollers.
- It is a large through-hole module rather than a component you can place in your own PCB layout, and it eats a ridiculous amount of space inside an enclosure.
- There is no temperature compensation, and since the speed of sound changes with temperature, your distance readings drift with the weather.
- Condensation, fragile exposed transducers, and the acoustic changes caused by an enclosure or grille all shift your readings even further.

SWAP IT FOR

REPLACEMENT PART	WHY THIS PART
ST VL53 family (VL53L1X, VL53L4CD)	These time-of-flight sensors measure how long it takes light to bounce back from a target, and they are excellent for short-range distance sensing.
MaxBotix ultrasonic sensors	These product-grade sensors are not necessarily cheap, but their spec sheets and reliability are exactly what you are paying for.
Reflective IR sensor	A simple reflective IR sensor does the job for pennies if all you need is basic presence detection.

> Ultrasonic sensing itself is a perfectly valid approach for products, and the problem is this specific two-dollar module.

#5 HC-SR501 PIR Module

MOTION DETECTION

WHY IT FAILS IN A PRODUCT

- Sensitivity and timing are set with two screw-adjusted trimpots, and there is no way to replicate those settings consistently across thousands of units in production.
- The retrigger behavior is inconsistent, the onboard regulator has its own quirks, and false triggers are a constant complaint, especially with a Wi-Fi antenna nearby.
- In a security or lighting product, every false trigger chips away at your customer's trust until they stop using the product completely.
- Once again, this is a module with a form factor that has no place on a production board.

SWAP IT FOR

REPLACEMENT PART	WHY THIS PART
Panasonic EKMC series	These sensors give you a clean digital output, specified sensitivity, a whole range of lens options, and a package designed to be soldered directly into products.
Panasonic EKMB series	The EKMB versions are made specifically for battery products, with current draw of just a few microamps.
ST STHS34PF80	This infrared sensor gives you true presence detection, so it still knows a person is there after they stop moving.

#4 MQ-Series Gas Sensors (MQ-2, MQ-135)

GAS AND AIR QUALITY

WHY IT FAILS IN A PRODUCT

- The heater inside these sensors burns roughly 750 to 900mW continuously, so a battery-powered product is dead on arrival.
- They need a burn-in period of a day or more before readings settle, they drift constantly over their lifetime, and they respond to practically every gas at once.
- Without calibration against reference gases, which you cannot realistically do, the numbers coming out of these sensors mean almost nothing.
- You can never use these for any kind of safety function, because gas and carbon monoxide detection is a certified category with strict standards, and that liability lands directly on you.

SWAP IT FOR

REPLACEMENT PART	WHY THIS PART
Sensirion SGP40 / SGP41	These parts measure volatile organic compounds, the airborne chemicals given off by things like paints and cleaners, and they are the right choice for general air quality sensing.
Sensirion SCD40 / SCD41	These parts give you real CO2 measurement in a tiny package.
SPEC Sensors and Figaro electrochemical cells	An electrochemical cell is the way to detect one specific target gas.
Certified detection modules	A certified module costs more, but it comes with the testing and documentation that regulators, retailers, and insurance companies demand for any safety claim.

#3 MPU-6050 IMU

MOTION TRACKING

WHY IT FAILS IN A PRODUCT

- The MPU-6050 actually works pretty well, and that is exactly what makes this one sneaky.
- TDK InvenSense officially flagged the part as not recommended for new designs years ago, and the part is discontinued.
- A lot of what is sold on hobby modules today is remarked or flat-out counterfeit silicon, so every new reel can behave a little differently, and that turns your firmware tuning into a guessing game.
- You cannot build a production product around a discontinued part, because you are one supply hiccup away from a stopped production line.

SWAP IT FOR

REPLACEMENT PART	WHY THIS PART
TDK ICM-42688-P	This is the modern successor from the same company that made the MPU-6050, with better specs and lower power.
Bosch BMI270	This part is optimized for low power, which makes it a strong choice for wearables and battery products.
ST LSM6 family (such as the LSM6DSO)	This family gives you a wide range of options with mature software support, and all of them are actually in production.

A warning from the real world

One Hardware Academy member built his design around the 9-axis MPU-9250 and had to switch parts mid-development when it went end of life. The swap was not trivial, because the replacement ran its data lines at 1.8V, so he had to add level shifting just to talk to the new part. When a manufacturer marks a part as not recommended for new designs, that part is living on borrowed time.

#2 CdS Photoresistors (LDRs)

AMBIENT LIGHT

WHY IT FAILS IN A PRODUCT

- The classic photoresistor is made from cadmium sulfide, and cadmium is restricted under RoHS, the European regulation that limits hazardous substances in electronics.
- In practice, that means these photoresistors cannot go into a product sold in the EU, so the most familiar light sensor on Earth is effectively banned from one of the biggest markets in the world.
- Even if compliance were not a problem, they respond slowly, nobody fully specifies which wavelengths they react to, and the part-to-part variance is so wide they barely qualify as sensors.

SWAP IT FOR

REPLACEMENT PART	WHY THIS PART
Vishay VEML7700	This sensor is a popular first choice with a wide dynamic range.
TI OPT3001	This sensor is designed to match the response of the human eye.
Lite-On LTR-303	This sensor is a compact, low-cost option.

All three give you calibrated brightness readings in lux over I^2C, they are fully RoHS compliant, and in volume they cost around a dollar or less.

#1 ACS712 Modules on Mains Voltage

AC CURRENT SENSING

WHY IT FAILS IN A PRODUCT

- The Allegro chip itself is a legitimate part, but the hobby breakout board it sits on is not.
- These modules route mains voltage through PCB traces with nowhere near the creepage and clearance that safety standards require, and that is just the minimum spacing between high voltage and everything else on the board.
- There is no reinforced isolation, and the screw terminals are often completely exposed.
- Inside a product, that is a shock hazard, a fire hazard, and an automatic certification failure all rolled into one.

SWAP IT FOR

REPLACEMENT PART	WHY THIS PART
Shunt resistor + TI AMC1302	A shunt paired with an isolated amplifier, laid out on your own board with the proper spacing, is the standard approach.
Current transformer	A current transformer clamps around a wire and gives you an isolated measurement by design, which keeps mains off your board entirely.
Allegro ACS37800	This newer part is made to be integrated correctly into your own layout.
Certified energy metering modules	A certified metering module is the safe option when you would rather buy the safety engineering instead of designing it.

Safety warning

Treat anything that touches mains voltage with the respect it deserves, because this is the one category where a bad sensor choice can actually hurt somebody.

The Five Production Filters

Once you move from a bench prototype to a manufacturable product, every sensor gets judged on the same five filters, and a part has to clear all of them. Run every component in your next design through this list before you commit to a layout.

1 **Accuracy**
The datasheet must specify performance across your full operating range, not just at a comfortable room temperature.

2 **Lifecycle**
The part must be in active production, because anything marked as not recommended for new designs is living on borrowed time.

3 **Compliance**
Every substance in the part must clear RoHS and the regulations of every market where you plan to sell.

4 **Power**
The sensor has to fit inside your power budget, and that matters most in battery products.

5 **Safety**
Anything that touches mains voltage or makes a safety claim needs proper isolation, spacing, and certification behind it.

Get a second opinion before it ends up in your design.

Component choices like these are an everyday conversation inside the Hardware Academy, where you can get feedback on your specific product from engineers who have taken real designs through production.

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